Cybersecurity Data Analysis

The Weakest Link in Cybersecurity

- Security systems are only as strong as their weakest component（木桶效應）
- In many attacks the weakest component is the human user（人是軟弱的）
- Attackers manipulate people instead of hacking systems（所以從人下手）

What is Social Engineering

- Psychological manipulation to obtain confidential information
- Targets human trust rather than technical vulnerabilities
- Often the first step in large cyber attacks

Social Engineering Attack Lifecycle

1. Recon
2. Target
3. Trust
4. Expliot
5. Escalate

Psychological Triggers Used by Attackers

1. Authority：假裝自己是頂黑客師傅
2. Urgency：用時間緊急上壓力給對方，不要讓他多想
3. Fear：就抓住對方弱點威脅
4. Curiosity：澳門合關線上發排免費玩
5. Trust：impersonating（冒充）colleagues（同事）

Human-Based Social Engineering

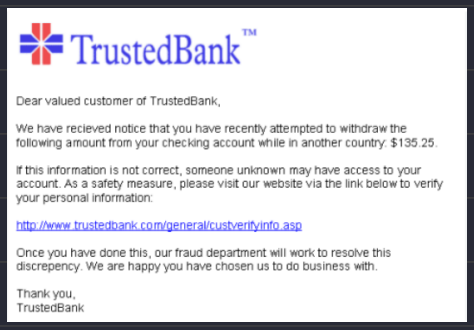
1. Shoulder surfing：肩膀後面凝視，看密碼
2. Dumpster diving：翻實體垃圾桶，看有沒有機密文件
3. Tailgating：尾隨他人，對方不知情
4. Piggybacking：搭便車，要對方幫你開門，對方知情
5. Impersonation (Vishing)：冒充，假裝自己是別人

Mobile-Based Attacks

1. SMiShing (SMS phishing)：
2. Malicious mobile apps
3. Repackaged apps
4. Fake security applications

Computer-Based Attacks

1. Phishing emails: 攻擊者偽裝成值得信任的實體，發送電子郵件誘騙受害者


TrustedBank™

Dear valued customer of TrustedBank,

We have recieved notice that you have recently attempted to withdraw the following amount from your checking account while in another country: $135.25.

If this information is not correct, someone unknown may have access to your account. As a safety measure, please visit our website via the link below to verify your personal information:

http://www.trustedbank.com/general/custverifyinfo.asp

Once you have done this, our fraud department will work to resolve this discrepency. We are happy you have chosen us to do business with.

Thank you,
TrustedBank

   a. Spear Phishing （魚叉式釣魚）：位目標量身打造釣魚郵件，例如要你做 Q3 財報
   b. Whaling （捕鯨攻擊）：專攻 CEO 或高管，要玩只玩大的
   c. Pharming (網址嫁接 / 網域欺騙)：轉發 (Redirect) 網路請求到攻擊者網站
   d. Spimming (即時通訊垃圾攻擊)：Spam (垃圾訊息) + IM (Instant Messaging 即時通訊)
2. Fake login pages：架設一個與官方很像的登入頁面，但密碼直接傳給攻擊者
3. Spam campaigns (垃圾郵件活動)：未經請求就大量發送的電子郵件
4. Scareware：以掃描到 2000+ 病毒，立即下載防毒軟體
5. Pop-up attacks：在你瀏覽網頁時，強行跳出在當前視窗之上的額外瀏覽器視窗。